DIY Activity: Product Recall Data Analysis and Insights

Guidelines for Students: *This is a self-paced Do-It-Yourself (DIY) activity and* ***submissions are not required****. The primary objective is to enhance your skills through practical, hands-on learning. Engage in discussions with peers to deepen your understanding and collaboratively work through the activity. If you have any questions or need clarification,* ***please raise them during the upcoming Live Sessions*** *for a more comprehensive explanation.*

Overview: *You are tasked with analyzing product recall data to identify high-risk products by examining Defect Rate, Units Recalled, and Reasons for Recall. The goal is to categorize defect severity and uncover trends that will help you make informed decisions regarding product quality, risk management, and compliance.*

Dataset:

product_recall dataset

- ***Product ID-*** *Unique identifier for each product.*
- ***Batch Number-*** *Identifier for the production batch of the product.*
- ***Manufacturing Date-*** *Date the product was manufactured.*
- ***Recall Date-*** *Date the product was recalled. Recall is defined as the product retrieval due to defects*
- ***Defect Rate-*** *Proportion indicating the rate of defects in the product. To convert it to a percentage, you would multiply the value by 100.*
- ***Units Recalled-****The total number of units recalled for the product.*
- ***Reason for Recall-****The reason specified for the product's recall.*

Tasks to be performed:

- *Use CONDITIONAL FORMATTING with a CUSTOM formula that utilizes the AND(logical_expression1, [logical_expression2, ...]) function to highlight rows where the 'Defect Rate' exceeds 15% and the 'Units Recalled' are greater than 500. After highlighting, determine how many such rows exist, and use FILTERS to find the most common reason for recall among these highlighted rows.*

- *Calculate the 'Total Defect Impact' by multiplying the 'Defect Rate' and 'Units Recalled'. What is the average Total Defect Impact across all products?*

- *Create a formula with absolute REFERENCING to calculate penalties for recalls, where the penalty is determined by multiplying the 'Units Recalled' by a fixed penalty*

rate of 2, which is stored in a specific cell. What is the total penalty for all recalls based on this calculation?

- *Use the 'Reason for Recall' column to count the number of recalls for each unique reason. Which reason has the highest number of recalls?*

- *Find the earliest and latest Recall Date in the dataset. What is the time gap in days between the two dates?*

- *Determine the number of products with a 'Defect Rate' between 10% and 20% by using filters.*

- *Combine the 'Product ID' and 'Batch Number' into a single identifier for each product using the '&' OPERATOR with a '-' in between the two values. What is the identifier for the first product in the dataset?*

- *Identify products where the Defect Rate is in the top 10% of the dataset. Highlight these rows and calculate the total Units Recalled for these products.*

- *Classify the 'Defect Rate' into the following categories using the CONDITIONAL FORMATTING: Low for a Defect Rate less than or equal to 10%, Medium for a Defect Rate greater than 10% and less than or equal to 20%, and High for a Defect Rate greater than 20%. Then, determine how many products fall into each category using FILTERS.*

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